

CS & IT ENGINEERING



'C' Programming

'C' TOKENS

Lecture No.- 03



By- Satya sir

Recap of Previous Lecture



- 'C' Token ?

- Identifiers
- keywords
- Constants
- Strings

- Structure of a 'C' Program

- Comments



Topics to be Covered



- Operators^{***}

- Types

- Unary operators





Topic : 'C' Tokens - 3



Operator ? Which Performs operation on/using operands.

Ex: $a * b$

a, b operands

$*$ operator

Multiplication operation

- Based on Number of operands, that can be used at a time, operators are classified into 3 types:

- 1) Unary operators $\longrightarrow$ Single operand
- 2) Binary operators $\longrightarrow$ Two operands
- 3) Ternary operator $\longrightarrow$ Three operands

4) Special operators

* $(), \{ \}, [] \Rightarrow$ Brackets

* $\rightarrow, \cdot \Rightarrow$ member access operators

* $, (comma) \Rightarrow$ Comma operator



Topic : 'C' Tokens - 3

Unary operators

<u>Operator</u>	<u>Operation</u>
++	Increment
--	Decrement
+	Sign Plus
-	Sign minus
*	Pointer Indirection
&	Address of
sizeof	Size of
!	Logical NOT/Negation
~	Bitwise NOT/Complement

Binary operators

1) Arithmetic

<u>operator</u>	<u>Operation</u>
+	Addition
-	Subtraction
*	Multiplication
/	Division
./	Modulus

<u>Logical Operator</u>	<u>operation</u>
&&	Logical AND
	Logical OR

Assignment operator

=

Bitwise and shift

<u>Operator</u>	<u>operation</u>
&	Bitwise AND
	Bitwise OR
^	Bitwise XOR
<<	Left shift
>>	Right shift

Relational (or) Comparison

<u>operator</u>	<u>operation</u>
<	Less than
<=	Less than or Equal to
>	Greater than
>=	Greater than or Equal to
!=	Not Equal to
==	Equal to

Ternary operator

?:





Increment (++) , Decrement (--) operators

$++ \Rightarrow +1$ $-- \Rightarrow -1$

The ++, -- operators can be used in either of 2 forms:

① Prefix form

- <u>Pre Increment</u>	++ operand	<u>Ex:</u> ++i, ++x
- <u>Pre Decrement</u>	-- operand	--i, --x...

Increment/Decrement before Assignment

② Postfix form

- Post Increment : operand ++	<u>Ex:</u> i++, x++
- Post-Decrement : operand --	i--, x--

Increment (or) Decrement after Assignment



Topic : 'C' Tokens - 3



Examples :

① $a=3, b, c, d, e$

$b = ++a;$

$c = b++;$

$d = --c;$

$e = d--;$

$a = \underline{4} \quad b = \underline{5} \quad c = \underline{3}$

$d = \underline{2} \quad e = \underline{3}$

a $\boxed{\cancel{3}4}$

b $\boxed{\cancel{4}5}$

c $\boxed{\cancel{4}3}$

d $\boxed{\cancel{3}2}$

e $\boxed{3}$

$b = ++a$

$\boxed{a = a + 1}$
 $b = a$

$a = 4, b = 4$

$c = b++$

$\boxed{c = b}$
 $b = b + 1$

$b = 5, c = 4$

$d = --c$

$\boxed{c = c - 1}$
 $d = c$

$c = 3, d = 3$

$e = d--$

$\boxed{e = d}$
 $d = d - 1$

$d = 2, e = 3$



Topic : 'C' Tokens - 3



Ex:

(2)

$a = 3, b = 4, c, d, e$

a	b	c	d	e
$\cancel{3}/4$	$\cancel{4}/3$	$\cancel{4}/6$ 7	$\cancel{-2}/-3$	3

$c = a++ + --b \quad c = 3 + 4 = 7$

$d = b-- - --c \quad d = 4 - 6 = -2$

$e = c++ + --d \quad e = 6 + (-3) = 3$

$a = \underline{4} \quad b = \underline{3}$

$c = \underline{7} \quad d = \underline{-3}$

$e = \underline{3}$

A) 4, 3, 7, 0, 6

B) 4, 3, 8, 10, 17

C) 4, 3, 7, -4, 2

D) 4, 3, 7, 3, 3

☒ E) 4, 3, 7, -3, 3

F) 4, 4, 9, -4, 3

G) 4, 3, 7, 0, 7

H) 3, 4, 6, -3, 9



Topic : 'C' Tokens - 3



Ex: 3

$$P=7, q=-P, r, s, t$$

$$r = P++ + ++q \quad r = 7 + (-6) = 1$$

$$s = P-- + q++ - --r \quad s = 8 + (-6) - 0 = 2$$

$$t = --P - --q + r++ + ++s \quad t = 6 - (-6) + 0 + 3 = 15$$

$$P = \underline{6} \quad q = \underline{-6} \quad r = \underline{1} \quad s = \underline{3} \quad t = \underline{15}$$

$$\begin{array}{c} P \\ \boxed{7} \\ \cancel{7} \\ 6 \end{array}$$

$$\begin{array}{c} q \\ \boxed{-7} \\ \cancel{-7} \\ -6 \end{array}$$

$$\begin{array}{c} r \\ \boxed{8} \\ \cancel{8} \\ 1 \end{array}$$

$$\begin{array}{c} s \\ \boxed{7} \\ \cancel{7} \\ 3 \end{array}$$

$$\begin{array}{c} t \\ \boxed{15} \end{array}$$



Topic : 'C' Tokens - 3



Ex: 4 [Conflict Example]

$$i = 10, j$$

$$j = ++i - i++$$

NOTE: The behavior of $++$, $--$ operators, when performed on single variable in the same expression depends on compiler / highly unpredictable.

Compiler - X

$$1) ++i \Rightarrow i = 11$$

$$2) i++ \Rightarrow \text{Assign } 11, i = 12$$

$$j = 11 - 11$$

$$i = 12, j = 0$$

Compiler - Y

$$1) i++ \Rightarrow \text{Assign } 10, i = 11$$

$$2) ++i \Rightarrow i = 12$$

$$j = 12 - 10 \\ = 2$$

$$i = 12, j = 2$$

Compiler - Z

$$++i = i = 11$$

$$i++ \Rightarrow \text{Assign first} \Rightarrow \text{Assign } 11$$

$$j = 11 - 11 = 0$$

After assigning 0 to j , Now Increment on ' i '.

$$i = 12, j = 0$$



2 mins Summary



- operator ?
 - List of operators
 - Unary operators
- $++$, $--$

To be Contd ... 



THANK - YOU